

EQUITY DATA ANALYSIS REPORT

1. Executive Summary

Objective

The objective of this analysis was to identify the five best-performing securities over the period:

01 June 2021 to 13 October 2021 based on their closing price appreciation.

For the selected stocks, the analysis also calculated:

- Percentage performance over the period.
- 30-day average trading volume as of 13 October 2021.
- Historical price evolution.
- Relative performance using normalized prices.

The analysis was performed using the provided historical equity dataset containing daily market information for 100 companies.

2. Approach and Methodology Note

Data loading and Preparation

The provided dataset was imported into Python using the Pandas library. The dataset contained daily market data for 100 companies, including the following metrics:

Column	Description
date	Trading date
isin	Security identifier
ticker	Company ticker
name	Company name
open	Opening price
high	Daily high price
low	Daily low price
close	Closing price
volume	Daily trading volume

Dataset Size:

Companies analyzed: **100**

Trading observations: **26,100**

Analysis period:

Also note that the date column was converted from string format into a datetime format to allow accurate filtering and chronological sorting.

Before: "2021-06-01" After: 2021-06-01 00:00:00

Before analysis, I equally inspected the dataset structure, checked data types, identified missing values, and verified the number of observations per company.

- Data was sorted by 'Ticker' and 'Date' to ensure that the first and last observation corresponds to the beginning and end of the analysis period respectively.
- The dataset contained missing values in the price and volume columns for a small number of records.
- These incomplete records were removed before performing calculations to avoid inaccurate performance and volume metrics.

3. Performance Calculation

The performance for each security was calculated as:

$$\text{Performance (\%)} = \frac{P_{\text{end}} - P_{\text{start}}}{P_{\text{start}}} \times 100$$

where:

- P_{start} = closing price on 2021-06-01
- P_{end} = closing price on 2021-10-13

Example:

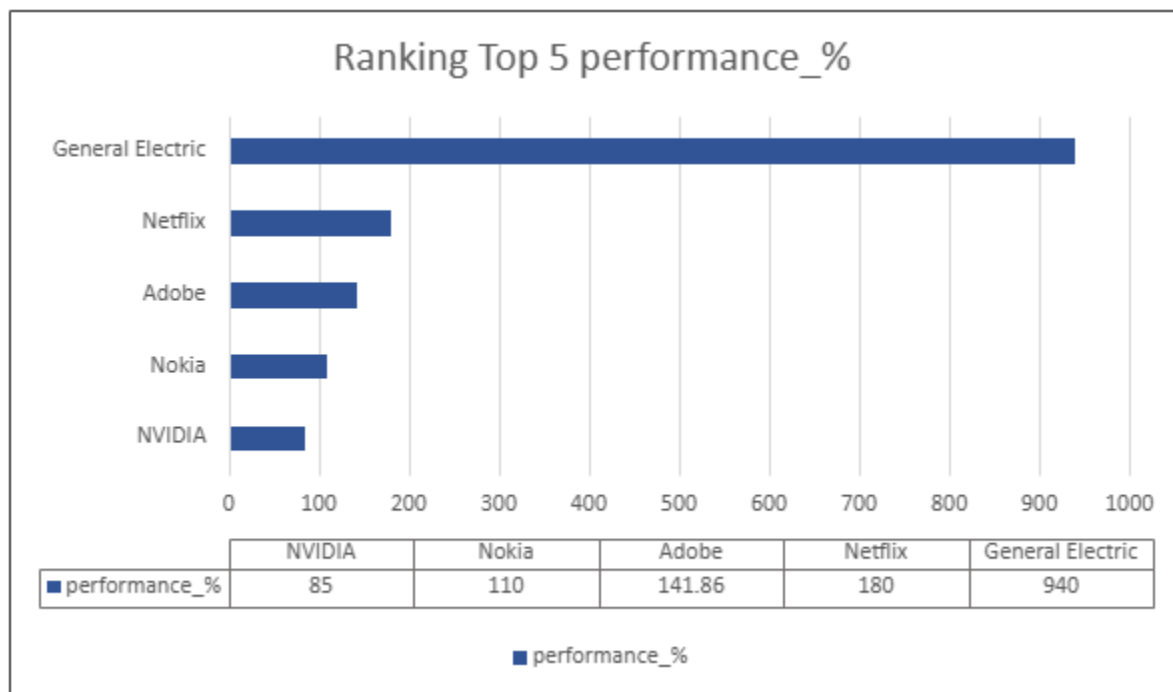
$$\begin{aligned} \text{General Electric, Performance (\%)} &= \frac{204.5057 - 19.6640}{19.6640} \times 100 \\ &= 940.0 \% \end{aligned}$$

4. Top 5 performing Titles

Results Summary table

Rank	Company	Ticker	ISIN	Performance	30-Day Avg Volume
1	General Electric	GENE	XX00000000010	940.00%	8.17M
2	Netflix	NETF	XX00000000028	180.00%	4.97M
3	Adobe	ADOB	XX00000000036	141.86%	3.11M
4	Nokia	NOKI	XX00000000044	110.01%	5.13K
5	NVIDIA	NVID	XX00000000051	85.00%	8.28M

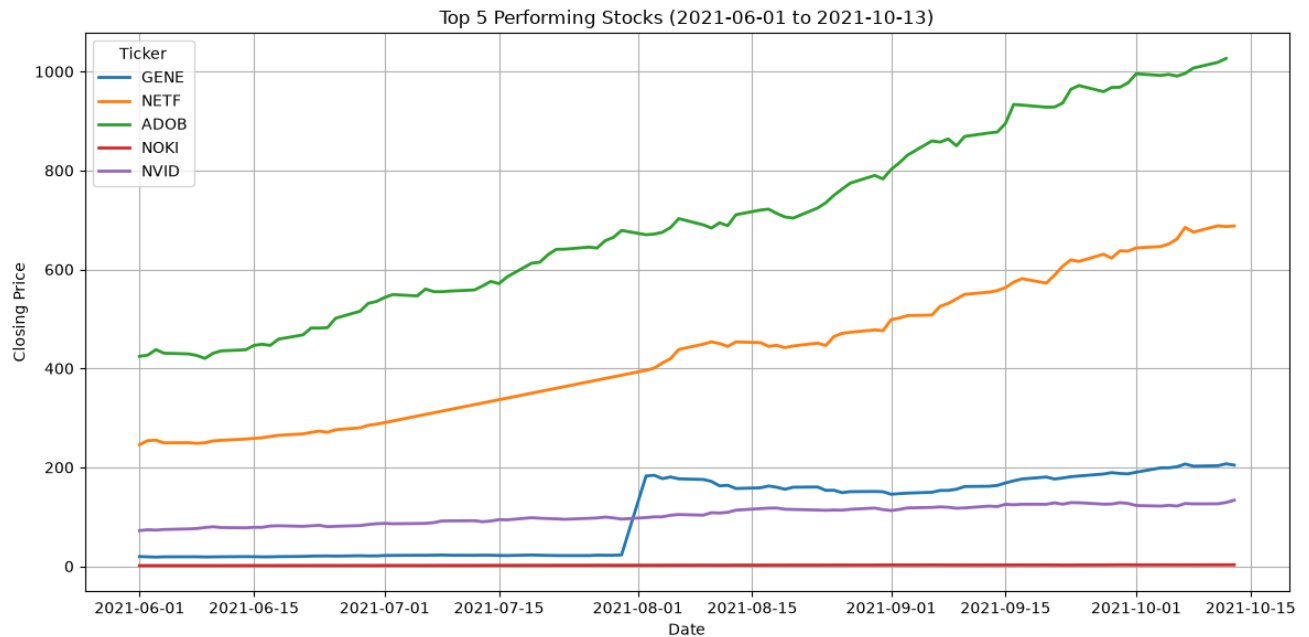
5. Performance Ranking Visualization



The bar chart provides a direct comparison of cumulative performance across the selected securities.

The strongest performer was **General Electric**, which showed a significantly larger price appreciation compared with the remaining equity securities.

6. Prices Visualization of Top performing securities



plot allows observation of:

- Price trends.
- Volatility.
- Timing of major price movements.

However, because securities have different initial price levels, direct comparison of absolute prices can be misleading. Therefore, the prices were normalized to ease this comparison of securities' performances.

Normalized Prices Analysis

The normalized chart provides a better comparison of investment performance because it removes the effect of different starting prices.

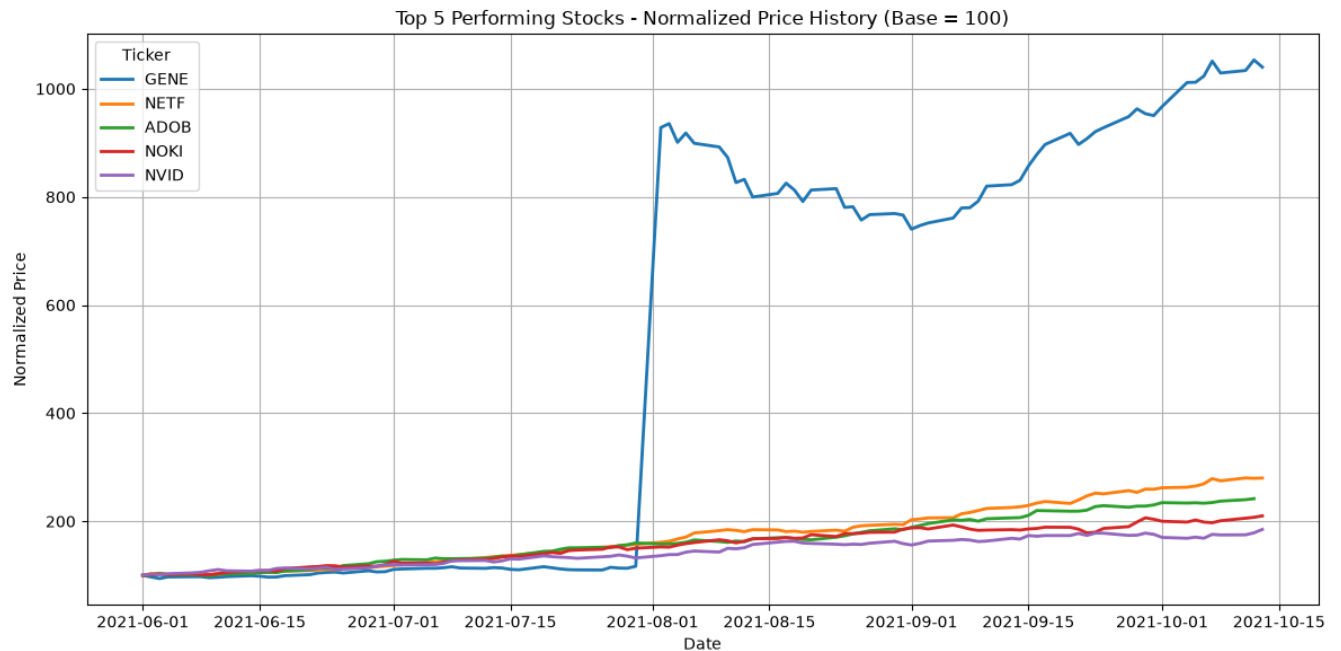
For example:

- A stock starting at €500 and increasing to €600 gained 20%. Meanwhile
- A stock starting at €10 and increasing to €30 gained 200%.

Raw prices alone will not reveal this difference. Each security was normalized using the formula;

$$\text{Normalized Price} = \frac{\text{Current Price}}{\text{Initial Price}} \times 100$$

Henceforth, all securities start at 100, and the final value represents relative growth



General Electric exhibited the strongest performance throughout the analysis period. The stock experienced relatively modest growth during June and July before undergoing a sharp upward movement at the beginning of August, where the normalized price increased rapidly from approximately **110** to over **900**. Although the price subsequently retraced and fluctuated during August and early September, it maintained substantially higher levels than the other securities before ending the period above 1,000. This corresponds to an overall gain of approximately **940%**, making General Electric the best-performing security in the dataset over the selected timeframe.

7. Conclusion

The analysis successfully identified the five strongest performing securities over the specified period and calculated their corresponding trading activity. The approach focused on

- Transparent calculations
- Reproducible methodology
- Appropriate handling of missing data
- Clear visualization of performance trends.

Additional normalized analysis was included to improve comparability between securities with different starting price levels.